Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

Sr.No. 8396

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:305 Course Section:1	Seat No: 2023701003

1_41_684_986_1_1505_41368

Name of the candidate. Rohit Mangale

Candidate's UID number: 2 0 2 3 7 0 1 0 0 3

Examination class room number: 3 0 5

Invigilator's signature with date: 8/10

(To be filled in by the candidate)

EXAMINATION: BE
(For example FY MCA/FY BTECH)

PROGRAM: CSE-DS

DAY AND DATE: Wednesday 08/10/25

COURSE NAME: DL

SEMESTER :

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

3338

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Q.2.b

A linear undercomplete autoencoder has ~~less~~ low compacity (i.e. is memory) so when the inputs are passed to the autoencoder n/w the n/w reduces the features / diemensions of the input.
~~Therefore in a under~~ latent space in a under complete autoencoder is smaller. ~~so~~
It is also ~~sot~~ good for feature extraction or diemension reduction, as it only keeps the important ~~th~~ structures of the data.
That is is why it is said a linear undercompete autoencoder trained with MSE learns a subspace equivalent to PCA.

An overcomplete autoencoder ~~ait~~ has a ~~too~~ compacity larger than the input data.
So when the input data is passed in the overcomplete autoencoder, the autoencoder just copies the ~~data~~ input data, and ~~it~~ not learns ~~properly regularized~~. the structures of the data.
~~If the model regularizes then the~~
Regularization techniques penalize the weights ~~the~~ so that the autoencoder does not ~~to~~ just copies the data but ~~extracts meaningful features~~ learns from the data.
Some regularization techniques are (L2) Ridge and (L1) Lasso which are used.

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Q1b. ~~The bias and variance~~ has high bias

~~Bias~~ :- A model is ~~bigged~~ when the model overfits the ~~a~~ training data and performances well in the trainging (has good training performance). But the model performs poorly in the testing as it does not generalize well and captures noise ~~to~~ from the training data.

The model has high variance when the model ~~overfits~~ underfits, performs poors in both training and testing.

The squared error penalizes large errors, which is higher for bias than variance.

When a models compacity is too high, the model learns the training data too well. So well that along with important features it also learns noise from the training data.
Therefore the model performs extremely well in training. It has a higher accuracy.
But in testing/validation, the model performs very poorly as it has not ~~generali~~ generalized well and not due to the noise captured, it performs bad.
It has higher ~~error~~ performance error in the testing/validation.
~~This case~~ Therefore when the model has a higher compacity it tends to overfit. This can be avoided using regularization or dropping.

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	Q2a.	

For a 2D convolution:

The no. of multiplications are.

$$f \times \left(\frac{H-f}{S}+1\right) \times f$$ [S is the strides]

For a depthwise separable convolution

$n_1 = \frac{H-f}{S}+1$

$n_2 = \frac{W-f}{S}+1$

[S is the no. of strides.]

$n_1 \times n_2 \times C \times f.$

Depthwise separable convolutions are critical ~~are~~ as they ~~understand / learn the patterns~~ extract the features from the images better. Due to performing ~~2~~ convolution separately on all the layers it helps to learn ~~&~~ more patterns and the n/w is able to perform well.

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Q3a.

The forget in LSTM works ~~fee~~ by letting the model forget (ie remove some information) ~~from~~ data ~~the we~~ and then the model weights are adjusted and this prevents the weights from decreasing and reducing to zero. The LSTM ~~has 4 gates~~ model can ~~lea~~ work with large sequences of data due to this, as it ~~can~~ remembers information and if the weights vanish or explode they are forgotten, and the merge gate latter ~~rem~~ & adds the previous output. which makes sure the information is not lost completely.

Thus the LSTM ~~works~~ prevents vanishing gradients under suitable initialization.

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